[0x1e]

BCD ex] $35_{10} = 0x35_{BCD}$

incorrect

~~cmp: al = 0x0a ⇒ ZF := 1~~

cmp: al < 0x0a ⇒ CF := 1

ie al < 10 ⇒ CF = 1 ELSE 0

sbb: al := al − 0x0a + CF

{0,1}

sub w/ borrow

$f(\text{hex val}) = \text{ASCII}$

BCD abuse

s

a valid packed BCD value

```
cmp     al,0x0a
sbb     al,0x69
das
```

ie al = al − 0x69

[0x1e]

lower-8 $\in$ rax

$\Downarrow$

cmp al, 0xa $\mapsto$ CF := 1 $\Leftrightarrow$ al < 0xa ~ 10 ~ 1001

~~else~~ CF := 0 iff al $\geq$ 0xa

sbb al, 0x69 $\mapsto$ al := al $-$ (0x69 + CF)

$\uparrow$ $\{0,1\}$

das $\mapsto$ decimal adjust AL after subtract

ie das(al, CF) $=$ $\begin{cases} al_0 - 6,\ AF := 1 & \left(\text{if } al \wedge 0x0f \ \underline{\text{or}}\ AF = 1\right) \\ al_0 - 0x60,\ CF := 1 & \left(\text{if } al > 0x9f \ \underline{\text{or}}\ CF = 1\right) \end{cases}$

$\underbrace{\quad}_{\text{flags}}$ $\underbrace{\quad}_{\text{conditions}}$

al $\in$ ~~[0,9]~~ [0,9] $\mapsto$ '0' $\to$ '9'

or

[a,f] $\mapsto$ 'A' $\to$ 'F'

$\Updownarrow$

Hex $\mapsto$ ASCI.

al

ex] f(0) = 0x30
f(1) = 0x31
f(2) = 0x32
⋮
f(9) = 0x39

$\longrightarrow$ 0 - 9 $\longmapsto$ 0x30 - 0x39
$\quad\quad\quad